



Python Library Functions





PYTHON LIBRARY FUNCTIONS

In this chapter, you'll learn about:

Python in-built functions

How to call such functions

Pass parameters to these functions

How to capture the result





PYTHON FUNCTIONS

There are three types of python functions:

- **Built-in functions**
 - Print, input, number and string functions
- **Library functions**
 - Math.Min(), math.Max(), statistics.Median()
- **User-defined functions**
 - Functions we write ourselves





PARAMETERS AND RETURN VALUE



Functions can take one or more parameter

- A value to be used in the function
- E.G.

```
print('Hello World!')
```

Return zero or one result

Do stuff!

- Many useful things
- Not there? Write it!



ABOUT FUNCTION PARAMETERS

A parameter can be...

- A literal

```
print('Hello World!')
```

- A variable

```
greeting = 'Hello World'  
print(greeting)
```

- An expression

```
print(100 * 0.2)
```





STANDARD LIBRARY FUNCTIONS



- **You've seen a few inbuilt functions**
 - Print, input, len, int, str, float, split
- **But there are many more**
 - Numeric functions
 - **Abs, min, max, pow, round**
 - String functions
 - **Capitalize, title, lower, upper**
 - **Zfill, format, ljust, rjust, center**
 - **Isdigit, startswith, endswith, replace**



BUILT-IN NUMERIC FUNCTIONS

```
numbers = [19,63,51,7,99,11,23,15,17,8]
```

```
print(min(numbers))
```

7

```
print(max(numbers))
```

99

```
print(pow(2,3))  
(or 2**3)
```

8

```
print(abs(-123))
```

123





ROUNDING FLOATS

```
print(round(5.671))
```

6

```
print(round(5.671,1))
```

5.7

```
print(round(5.671,2))
```

5.67

```
print(int(5.671))
```

5





ROUNDING FLOATS – MATH LIBRARY

```
import math
```

```
print( math.ceil(2000.98))
```

2001

```
print( math.floor(2000.98))
```

2000

`ceil(x)` the smallest integer $\geq x$

`floor(x)` the largest integer $\leq x$



FORMATTING STRINGS

```
str = "Bob"  
print(str.center(10, '-'))
```

.....▶

---Bob---

```
print(len(str))
```

.....▶

3

```
print(str.ljust(7)+"Smith")
```

...▶


Bob Smith

```
month="2"  
print(month.zfill(2))
```

.....▶

02





LOWERCASE AND UPPERCASE



```
str = "Bob"
```

```
print(str.lower())
```

bob

```
print(str.upper())
```

BOB

```
name = 'Bob'
if name == 'bob':
    print('Hello bob')
else:
    print("You're not bob!")
```

You're not bob!

```
if name.lower() == 'bob':
    print('Hello bob')
else:
    print("You're not bob!")
```

Hello bob

STRING FUNCTION EXAMPLES

```
str = "bob smith"
```

```
print(str.capitalize())
```

```
print(str.title())
```

```
print(str.replace(' ', '_'))
```

Bob smith
Bob Smith

bob_smith

```
print(str.title().replace(' ', '_'))
```

Bob_Smith

Can chain functions together

SPLIT FUNCTION AND IN COMMAND

Split a string into a List by a delimiter

```
cities = "london,birmingham,leeds,manchester,bristol"  
cityList = cities.split(',')  
cityList
```

```
london  
birmingham  
leeds  
manchester  
bristol
```

```
city = input('Please enter a city name: ')  
  
if city.lower() in cityList:  
    print('Your city is in the list!')
```

EXTRACTING PART OF A STRING



- You can extract part of a List

```
data = [1,3,5,7,9,11,13,15]  
print(data[1:5])
```

[3, 5, 7, 9]

- And that includes a String

```
word = 'abcdefgh'  
print(word[1:5])
```

bcde



TEST BEFORE CASTING TO INT

```
strAge = input('Please enter your age: ')\n\nif strAge.isdigit():\n    age = int(strAge)\n    print(age + 1)\nelse:\n    print(strAge, 'is not a valid age!')
```





STRING FORMAT FUNCTION

```
name="Bob"  
age=21  
city="London"
```

```
str = "{} lives in {}. He is {} years old".format(name,city,age)
```

```
print (str)
```

Bob lives in London. He is 21 years old

Press any key to continue ...



OTHER LIBRARIES



- There are hundreds of libraries
https://en.wikipedia.org/wiki/Category:Python_libraries
- Here are a few from the **statistics**

```
import statistics
```

Must import

```
numbers = [99,63,51,7,99,11,23,15,17,8]
```

```
print( statistics.mean(numbers) )           # average
```

```
print( statistics.median(numbers) )         # middle value
```

```
print( statistics.mode(numbers) )           # most common data
```



PYTHON FUNCTIONS

In this chapter, you learnt about:

- Python in-built functions
- Calling functions
- Pass parameters to these functions
- Capturing the result





Guided Task

- Please open the instructions in Cloud Academy



FURTHER READING

- <https://www.python.org/>
- https://www.tutorialspoint.com/python/python_strings.htm
- <https://docs.python.org/3/tutorial/index.html#tutorial-index>





Thank you!

Hope you enjoyed this learning journey!